MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, JK, DIGITAL, KEY CODE 6DE

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic JK flip-flop module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389 (except as modified in 3.4).

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:21Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 4 5 failures per million operating hours when predicted in accordance with MIL-STD-1389

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389 except the operating temperature range shall be 0° to 80°C.

3.5 Module configuration requirements The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein

3 5 1 Key code. The module key code shall be GDE.

3.5.2 Size. The module size shall be 1A

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5 6 Depth. The module shall have no additional depth requirements

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/2-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1.

3 7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3 8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5 0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4 QUALITY ASSURANCE PROVISIONS

4 1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665 Other equipment and components shall be as specified on the figures All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:21Z
Check the source to verify that this is the current version before use.

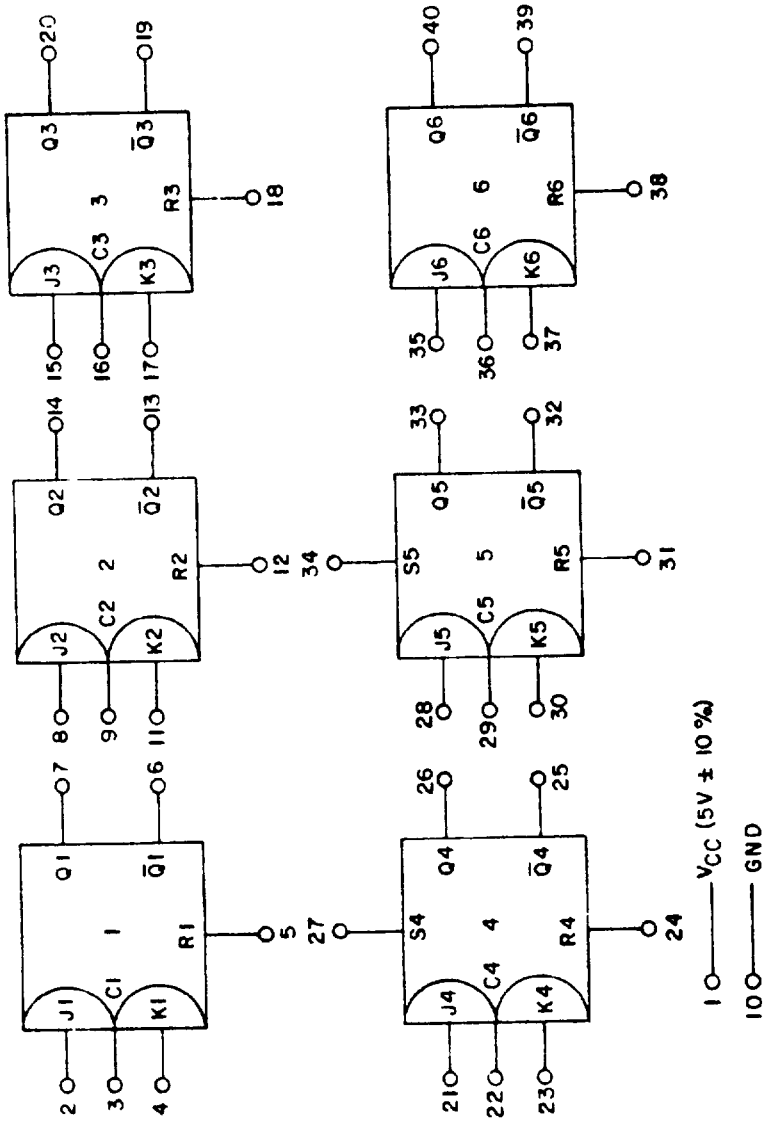


FIGURE 1. GDE module function diagram, JK flip-flop.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low loaded)	III	S	0 1	0.8	0.1	0.8	0.1	0.8	0.1	0 8	V
B1-B6	Input current (clock)	IV	S	0.0	1.0	0 0	1 0	0 0	1.0	0.C	1 0	mA
B7-B26	Input current (J, K, set and reset)	IV	S	0.0	0.5	0.0	0.5	0.0	0 5	0 0	0.5	mA
C1-C26	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1 5	-0.4	-1.5	-0.4	-1 5	V
D1-D6	Input current (clock), high level	VI	X	0	120	0	120	0	120	0	120	µA
D7-D26	Input current (J, K, set and reset), high level	VI	X	0	60	0	60	0	60	0	60	µA
E1-E8	Input current (clock), low level	VI	X	0.10	2.64	0.10	2 64	0 10	2.64	0 10	2 64	mA
E9-E20	Input current (J and K), low level	VI	X	0 10	1.73	0 10	1 73	0 10	1.73	0.10	1 73	mA
E21-E28	Input current (set and reset) low level	VI	X	0.10	2.64	0 10	2.64	0.10	2 64	0 10	2.64	mA
F1-F12	Output short circuit current, high level	VI	X	30	90	30	90	30	100	30	100	mA
G1-G6	Q output voltage, high level (J threshold, high level)	VII	X	2.4	4.5	2 4	4.5	2.4	4 5	2.4	4.5	V
H1-H6	$\overline{Q}$ output voltage, high level (K threshold, high level)	VII	X	2.4	4.5	2 4	4.5	2.4	4.5	2.4	4.5	V
J1, J2	Q output voltage, high level (set threshold, low level)	VII	X	2.4	4 5	2 4	4.5	2.4	4.5	2 4	4 5	V
J3-J8	$\overline{Q}$ output voltage, high level (reset threshold, low level)	VII	X	2.4	4.5	2.4	4 5	2.4	4 5	2.4	4 5	V
J9, J10	$\overline{Q}$ output voltage, high level (set threshold, high level)	VII	X	2.4	4 5	2 4	4.5	2.4	4.5	2.4	4 5	V
J11-J16	Q output voltage, high level (reset threshold, high level)	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2 4	4 5	V
J17, J18	$\overline{Q}$ output voltage, low level (set threshold, low level)	VII	X	0 00	0.40	0.00	0 40	0.00	0 40	0.00	0 40	V

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
J19-J24	Q output voltage, low level (reset threshold, low level)	VII	X	0.00	0.40	0.00	0.40	0.00	0.40	0.00	0.40	V
J25, J26	Q output voltage, low level (set threshold, high level)	VII	X	0.00	0.40	0.00	0 40	0.00	0.40	0.00	0.40	V
J27-J32	$\overline{Q}$ output voltage, low level (reset threshold, high level)	VII	X	0.00	0.40	0 00	0 40	0 00	0.40	0.00	0.40	V
K1-K6	Q output voltage, low level (J threshold, low level)	VII	X	0.00	0.40	0.00	0 40	0.00	0.40	0 00	0.40	V
K7-K12	$\overline{Q}$ output voltage, low level (K threshold, low level)	VII	X	0.00	0.40	0.00	0.40	0.00	0.40	0.00	0.40	V
L	Power supply current (Q, low level)	VIII	S	20	93	20	93	20	93	20	93	mA
M1-M6	JK operation (J and K threshold, high level, max freq, min pw)	IX	S	76	84	76	84	76	84	76	84	ns
M7-M12	JK operation (J and K threshold, low level; max freq, min pw)	IX	S	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
N1-N12	Transition time (min setup and hold time), t_{TLH}	X	S	3	20	3	20	3	25	3	25	ns
P1-P12	Transition time (min setup and hold time), t_{THL}	X	S	3	20	3	20	3	25	3	25	ns
Q1-Q12	Propagation delay time (min setup and hold time), t_{PHL}	XI	S	3	30	3	30	3	35	3	35	ns
R1-R12	Propagation delay time (min setup and hold time), t_{PLH}	XI	S	3	30	3	30	3	35	3	35	ns
S1, S2	Propagation delay time (t_{PHL}), set	XII	S	3	30	3	30	3	35	3	35	ns
T1, T2	Propagation delay time (t_{PLH}), set	XII	S	3	30	3	30	3	35	3	35	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
U1-U6	Propagation delay time (t_{PHL}), reset	XII	S	3	30	3	30	3	35	3	35	ns
V1-V6	Propagation delay time (t_{PLH}), reset	XII	S	3	30	3	30	3	35	3	35	ns
W1-W6	Maximum clock fall time	XIII	S	1.8	2.2	1.8	2.2	1.8	2.2	1 8	2.2	µs
X1-X6	Threshold clock input	XIII	S	76	84	76	84	76	84	76	84	ns
Y	Power supply transient rise time	XIV	S	2.5	10.0	2 5	10 0	2.5	10.0	2.5	1C 0	µs
Z	Isolation	XV	S	10	∞	10	∞	10	∞	10	∞	MΩ
AA1-AA12	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
BB1-BB12	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	mV
CC	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	mV
DD1-DD26	Input capacitance	XVIII	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

4.4.2 Order of testing. Test conditions A through DD shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through DD in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through Y, AA, and BB shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through Y, AA, and BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through DD shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/2 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658757. Part number M28787/2-1 supersedes part numbers 2658757 and M28787/2.

* 6.4 Module description. The following electrical parameters reflect the 0° to 80°C end-of-life limits from table I.

Supply voltage	5 Vdc ±10%
Maximum power dissipation (Q outputs at low level)	0.5 W (outputs not loaded)
Clock input	
Frequency	25 MHz maximum
Pulse width	20 ns minimum
Fall time	100 ns maximum
Voltage swing	1 8 V minimum

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

J and K input threshold voltage	
Low level	1.1 V minimum
High level	2.0 V minimum
Set, reset and clock input threshold voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Input current, low level (V_{IL} = 0 V)	
Clock	2.64 mA maximum
J and K	1 73 mA maximum
Set and reset	2.64 mA maximum
Input current, high level (V_{IH} = 2.4 V)	
Clock	120 µA maximum
J and K	60 µA maximum
Set and reset	60 µA maximum
Output voltage	
Low level	0.40 V maximum
High level	2.4 V minimum
Output drive current	
Low level	22 mA maximum
High level	1 32 mA maximum
Propagation delay time	
t_{PLH}	35 ns maximum
t_{PHL}	35 ns maximum
Set and reset delay time	35 ns maximum
Input setup time	0 ns minimum
Input hold time	0 ns minimum
Set and reset post time	15 ns maximum
Triggering edge	Negative

6.5 Module function. This module contains six independent JK flip-flops with both true (Q) and complementary ($\overline{Q}$) outputs. Each JK flip-flop has a J, K, clock, and reset input. JK flip-flop numbers 4 and 5 have an additional set input. Information is transferred into a temporary memory when the clock goes to a high state. When the clock returns low, the information is transferred to the Q and $\overline{Q}$ outputs. The information of the J and K inputs should be present when the clock goes high and the information should not be changed while the clock is high.

6.5 1 Input to output relationship. Each JK flip-flop has the input to output relationship of table II for positive logic.

6.5.2 Input-output. The application of a low state to the set will cause Q to go to the high state, application of a low state to the reset will cause Q to go to a low state. The clock must be in the low state when these functions are performed. The direct inputs can be applied at 15 ns (post time) after the clock signal has fallen. CAUTION If the preset or reset inputs are held at low while the J and K inputs are enabled, the "low" outputs will go high for approximately 20 ns on each excessive clock pulse.

6.5.3 Parallel outputs. Paralleling of outputs is not permissible.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:21Z
Check the source to verify that this is the current version before use.

6.6 Changes from previous issue The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

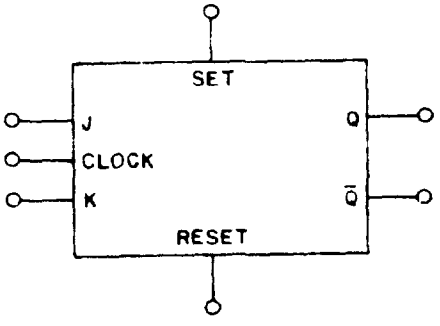
Agent
DLA - ES

Preparing activity.
Navy - EC

(Project 5963-N337)

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE II Input to output relationship.



J K Operation

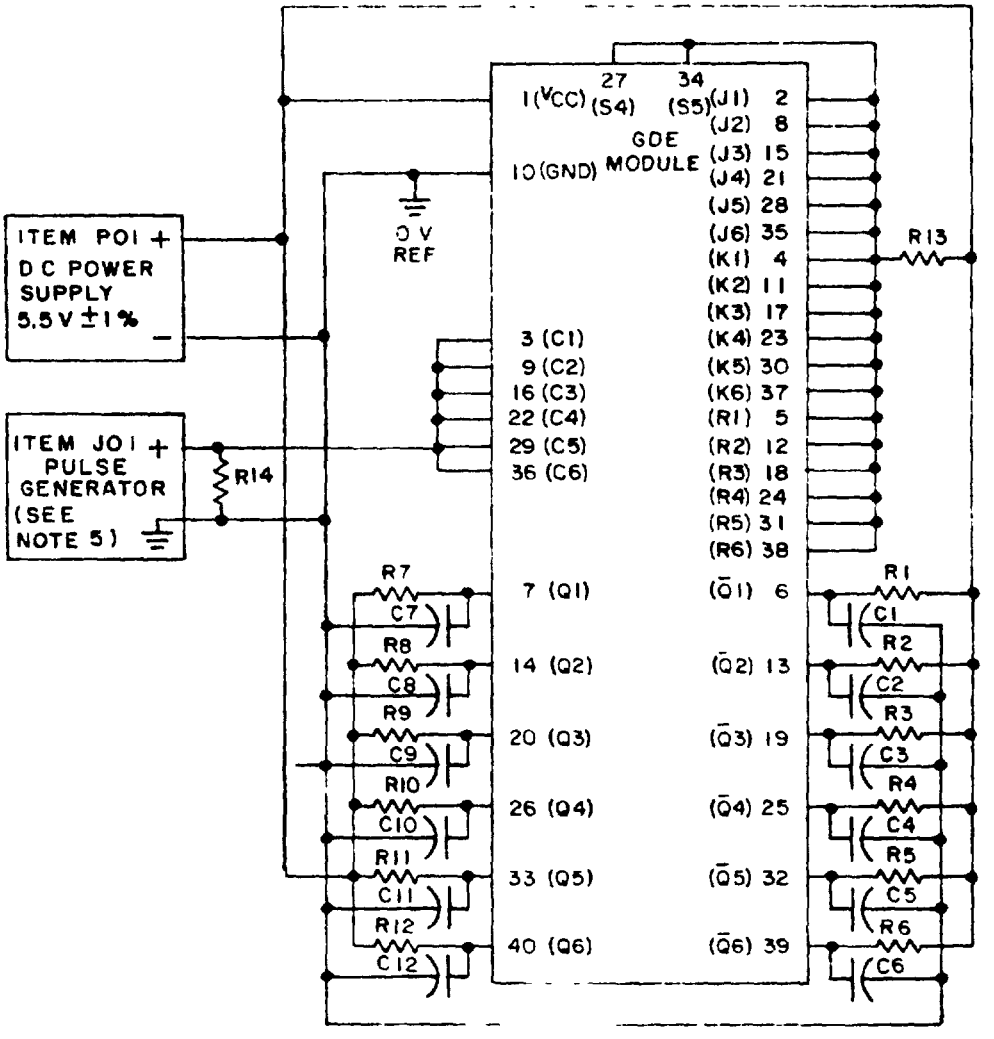
J	K	Qn	Qn+1
L	L	L	L
L	L	H	H
L	H	L	L
L	H	H	L
H	L	L	H
H	L	H	H
H	H	L	H
H	H	H	L

Asynchronous operation

Set	Reset	Qn
L	H	H
H	L	L

Qn = The nth output before clock pulse.
Qn+1 = The nth output after clock pulse.
L = Low level.
H = High level.

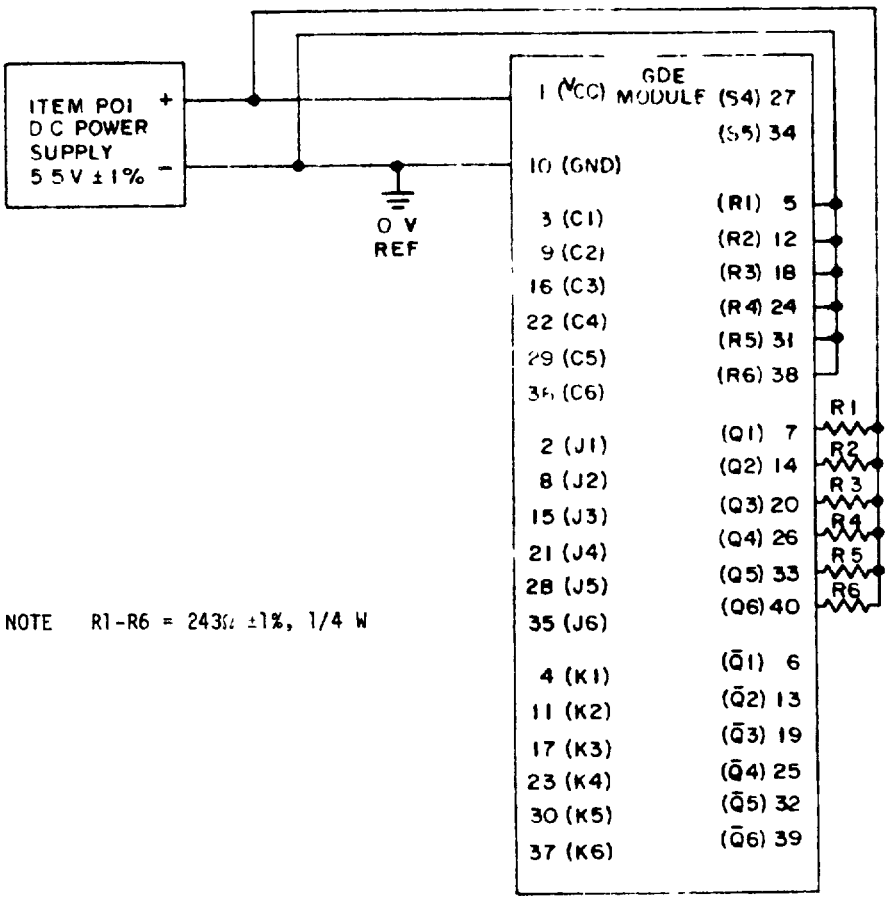
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES.
1. R1-R12 = 243Ω ±1%, 1/4 W
2. R13 = 1 kΩ ±1%, 1/8 W.
3. R14 = 50Ω ±5%, 1 W.
4. C1-C12 = 130 pF ±20%
5. Item J01 setup
   PA = 4 ±0.5 V.
   PW = 500 ±50 ns
   $t_{TLH} = t_{THL} = 25 \pm 5$ ns
   PRR = 1.0 ±0.1 MHz

* FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTE R1-R6 = 243Ω ±1%, 1/4 W

FIGURE 3 Component temperature test circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

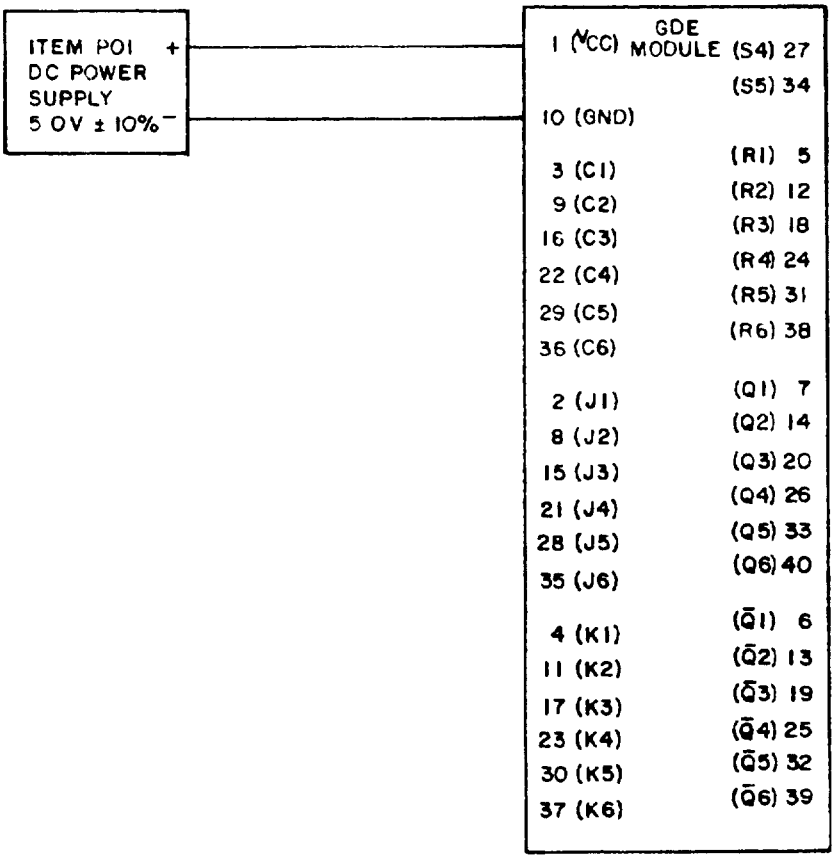


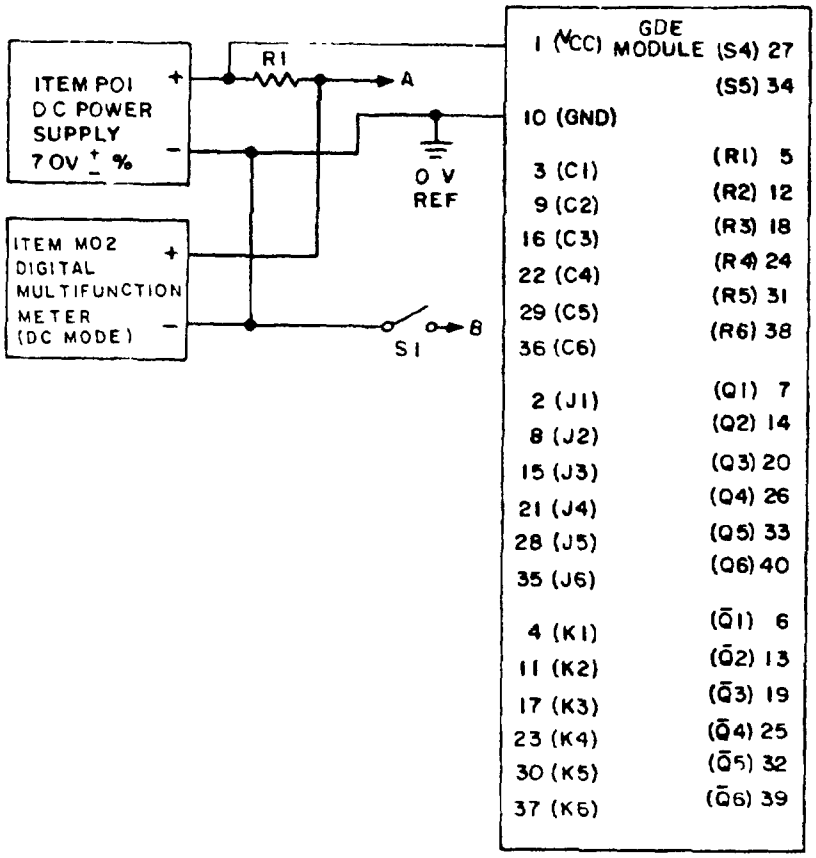
FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin 10 to pin	Min	Max		
A1	$\bar{Q}1$ V_{OUT} (low loaded)	6	3	4	0.1	0.8	V	5
A2	$\bar{Q}2$	13	9	11				
A3	$\bar{Q}3$	19	16	17				
A4	$\bar{Q}4$	25	Open	27				
A5	$\bar{Q}5$	32	Open	34				
A6	$\bar{Q}6$	39	36	37				
A7	Q1	7	Open	5				
A8	Q2	14		12				
A9	Q3	20		18				
A10	Q4	26		24				
A11	Q5	33		31				
A12	Q6	40		38				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES.
1. R1 = 243Ω ±1%, 1/4 W.
2. S1 = SPST switch.
3. Before connecting point "A", close switch S1 for a minimum of 100 ns, then open it.
4. After point "A" has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of five seconds before the measurement is recorded.

FIGURE 5 V_{OUT} (low-loaded) test circuit for table III

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE IV. Input current test.

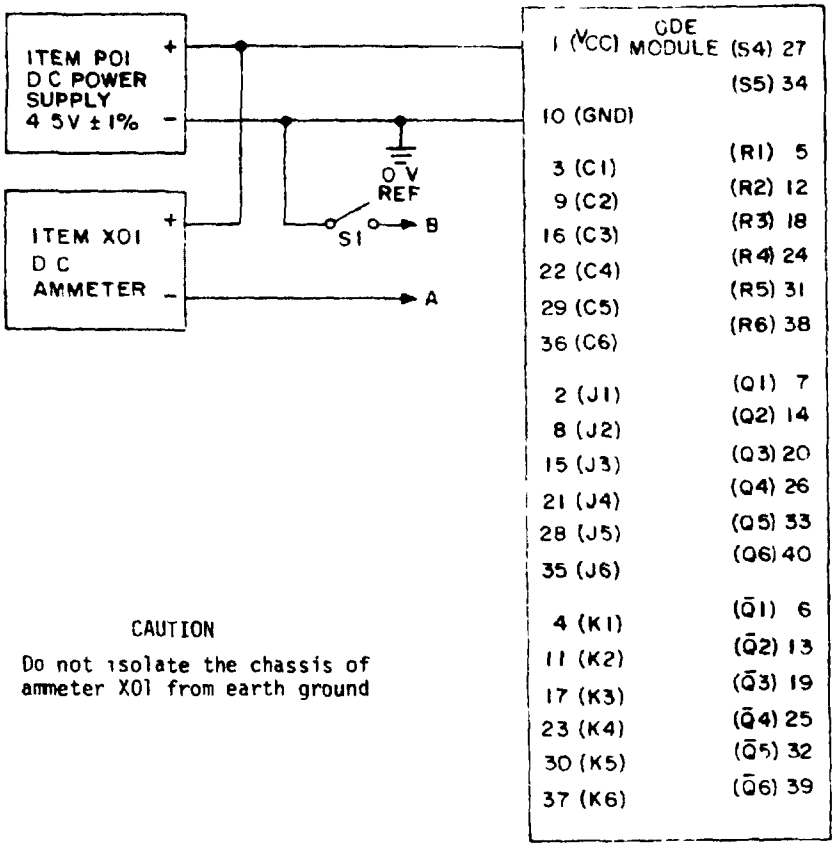
Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Pin 10 to pin	Min	Max		
B1	C1	Input current	3	Open	2, 4	0.0	1.0	mA	6
B2	C2		9		8, 11				
B3	C3		16		15, 17				
B4	C4		22		21, 23				
B5	C5		29		28, 30				
B6	C6		36		35, 37				
B7	J1		2		3		0.5		
B8	J2		8		9				
B9	J3		15		16				
B10	J4		21		22				
B11	J5		28		29				
B12	J6		35		36				
B13	K1		4		3				
B14	K2		11		9				
B15	K3		17		16				
B16	K4		23		22				
B17	K5		30		29				
B18	K6		37		36				

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE IV. Input current test - Continued.

Test condition	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin 10 to pin	Min	Max		
B19	R1 Input current	5	3	4	0.0	0.5	mA	6
B20	R2	12	9	11				
B21	R3	18	16	17				
B22	R4	24	Open	27				
B23	R5	31	Open	34				
B24	R6	38	36	37				
B25	S4	27	Open	24				
B26	S5	34	Open	31				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

NOTES:
1. S1 = SPST switch.
2. After making all circuit connections, close switch S1 for a minimum of 100 ns; then open it and make measurements.

FIGURE 6 Input current test circuit for table IV

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE V. Input clamp diode voltage test.

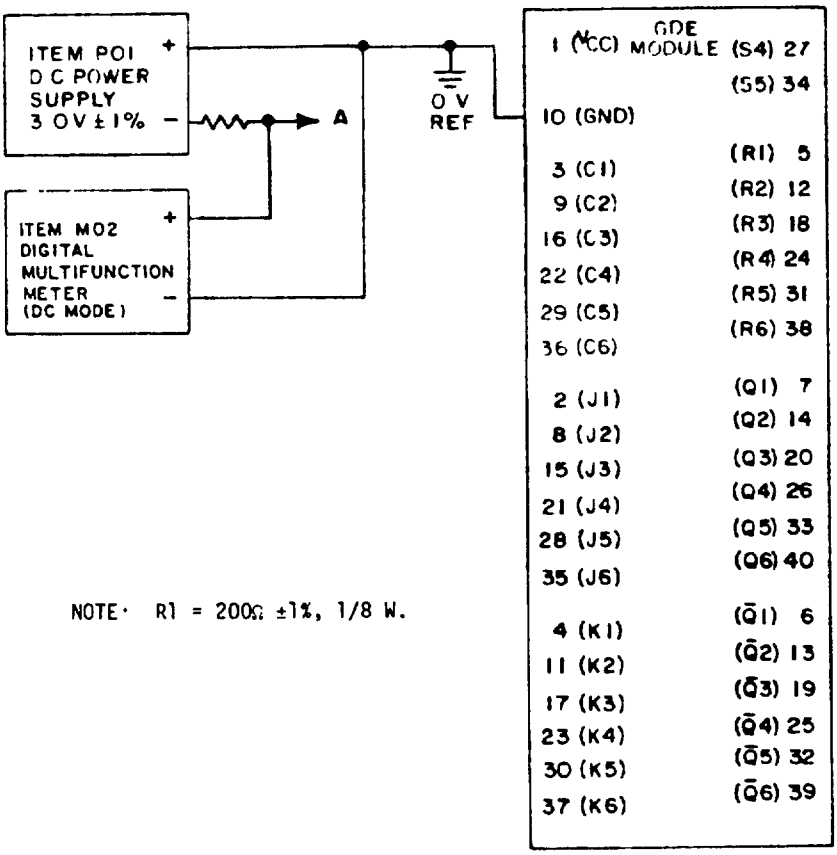
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	C1 Input clamp diode voltage	3	-0.4	-1.5	V	7
C2	C2	9				
C3	C3	16				
C4	C4	22				
C5	C5	29				
C6	C6	36				
C7	J1	2				
C8	J2	8				
C9	J3	15				
C10	J4	21				
C11	J5	28				
C12	J6	35				
C13	K1	4				
C14	K2	11				
C15	K3	17				
C16	K4	23				
C17	K5	30				
C18	K6	37				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE V. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C19	R1	Input clamp diode voltage	5	-0.4	-1.5	V	7
C20	R2	↓	12	↓	↓	↓	↓
C21	R3	↓	18	↓	↓	↓	↓
C22	R4	↓	24	↓	↓	↓	↓
C23	R5	↓	31	↓	↓	↓	↓
C24	R6	↓	38	↓	↓	↓	↓
C25	S4	↓	27	↓	↓	↓	↓
C26	S5	↓	34	↓	↓	↓	↓

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:21Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 200Ω ±1%, 1/8 W.

FIGURE 7 Input clamp diode voltage test circuit for table V

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE VI. Input current (high-level and low-level) and output short circuit current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	C1 Input current, high level	3	Open	Open	2,4	0	120	μA	
D2	C2	9			8,11				
D3	C3	16			15,17				
D4	C4	22			21,23				
D5	C5	29			28,30				
D6	C6	36			35,37				
D7	J1	2			3		60		
D8	J2	8			9				
D9	J3	15			16				
D10	J4	21			22				
D11	J5	28			29				
D12	J6	35			36				
D13	K1	4			3				
D14	K2	11			9				
D15	K3	17			16				
D16	K4	23			22				
D17	K5	30			29				
D18	K6	37			36				
D19	R1	5		3	4				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE VI. Input current (high level and low-level) and output short circuit current (high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
D20	R2	Input current, high level	12	Open	9	11	0	60	µA	8
D21	R3		18		16	17				
D22	R4		24		Open	27				
D23	R5		31		Open	34				
D24	R6		38		36	37				
D25	S4		27		Open	24				
D26	S5		34			31				
E1	C1	Input current, low level	2,4	3		5	0.10	2.64	mA	
E2	C2		8,11	9		12				
E3	C3		15,17	16		18				
E4	C4		21,23,27	22		24				
E5	C5		28,30,34	29		31				
E6	C6		35,37	36		38				
E7	C4		21,23,24	22		27				
E8	C5		28,30,31	29		34				
E9	J1		3	2		5		1.73		
E10	J2		9	8		12		1.73		

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE VI. Input current (high-level and low-level) and output short circuit current (high-level) test - Continued.

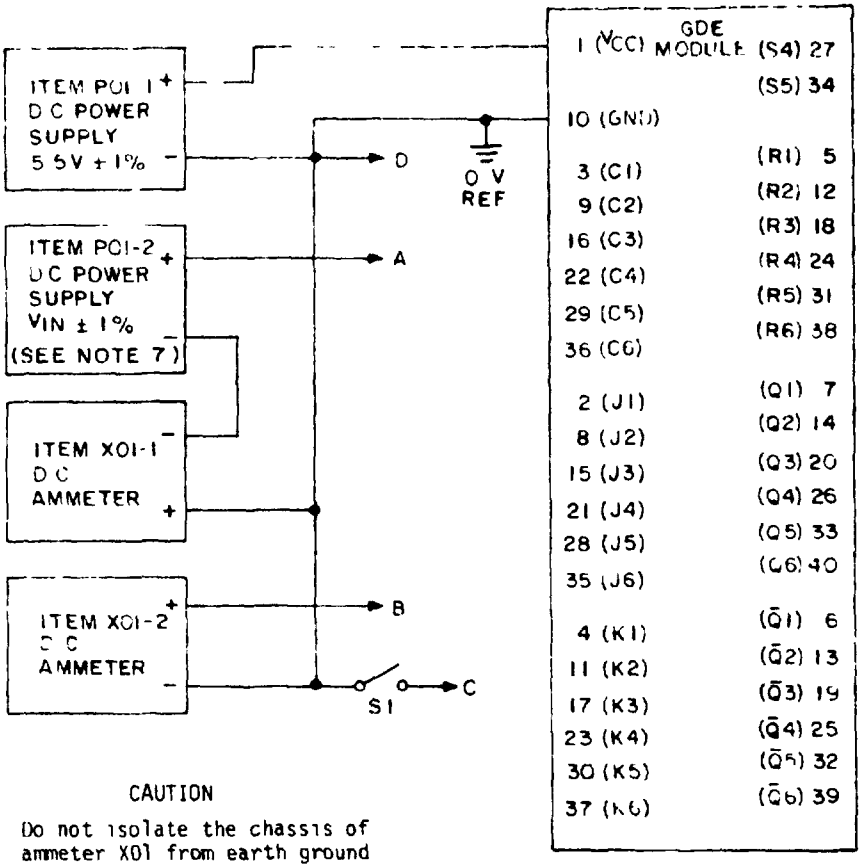
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E11	J3 Input current, low level	16	15	Open	18	0.10	1.73	mA	8
E12	J4	22	21		24				
E13	J5	29	28		31				
E14	J6	36	35		38				
E15	K1	3	4	3 (see note 6)	Open				
E16	K2	9	11	9 (see note 6)	Open				
E17	K3	16	17	16 (see note 6)	Open				
E18	K4	22	23	Open	27				
E19	K5	29	30	Open	34				
E20	K6	36	37	36 (see note 6)	Open				
E21	R1	2,4	5	Open			2.64		
E22	R2	8,11	12						
E23	R3	15,17	18						
E24	R4	21,23,27	24						

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE VI. Input current (high-level and low-level) and output short circuit current (high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
E25	R5	Input current, low level	28,30,34	31	Open	Open	0.10	2.64	mA	8
E26	R6		35,37	38						
E27	S4		21,23,24	27						
E28	S5		28,30,31	34						
F1	Q1	Output short circuit current (high level)	2	7	3	4	30	90		
F2	Q2		8	14	9	11				
F3	Q3		15	20	16	17				
F4	Q4		Open	26	Open	27				
F5	Q5		Open	33	Open	34				
F6	Q6		35	40	36	37				
F7	Q1		Open	6	Open	5				
F8	Q2			13		12				
F9	Q3			19		18				
F10	Q4			25		24				
F11	Q5			32		31				
F12	Q6			39		38				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

NOTES:
1. S1 = SPST switch.
2. Use item X01-1 for test condition D and X01-2 for test conditions E and F.
3. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
4. With point "A" disconnected, item X01-1 should indicate 0.0 current on the 1.0 µA range.
5. When performing output short circuit current test, only one output shall be shorted at a time, and the measurement taken within five seconds.
6. After making all circuit connections, close switch S1 for a minimum of 100 ns, then open it and make measurements. For test conditions F15, E16, E17, and E20, perform the switching required at point "C", then connect to point "A" and make measurement.
7. Adjust item P01-2 as follows
   V_{IN} = 2.4 V for test conditions D1-D26.
   V_{IN} = 4.5 V for test conditions E1-E28 and F1-F12.

* FIGURE 8. Input current (high-level and low-level) and output short circuit current (high-level) test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE VII. Output voltage (high-level and low-level) test. 1/

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G1	Q1 Output voltage, high level (J threshold, high level)	2	5	3	7	6	4	2.4	4.5	V	9
H1	$\overline{Q}$1 Output voltage, high level (K threshold, high level)	4	Open	3	6	7	2				
G2	Q2 Output voltage, high level (J threshold, high level)	8	12	9	14	13	11				
H2	$\overline{Q}$2 Output voltage, high level (K threshold, high level)	11	Open	9	13	14	8				
G3	Q3 Output voltage, high level (J threshold, high level)	15	18	16	20	19	17				
H3	$\overline{Q}$3 Output voltage, high level (K threshold, high level)	17	Open	16	19	20	15				
G4	Q4 Output voltage, high level (J threshold, high level)	21	24	22	26	25	23				
H4	$\overline{Q}$4 Output voltage, high level (K threshold, high level)	23	27	22	25	26	21				

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE VII. Output voltage (high-level and low-level) test - Continued. 1/

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
G5	Q_5 Output voltage, high level (J threshold, high level)	28	31	29	33	32	30	2.4	4.5	V	9
H5	$\overline{Q}_5$ Output voltage, high level (K threshold, high level)	30	34	29	32	33	28				
G6	Q_6 Output voltage, high level (J threshold, high level	35	38	36	40	39	37				
H6	$\overline{Q}_6$ Output voltage, high level (K threshold, high level)	37	Open	36	39	40	35				

-- footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE VII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
J1	Q4	Output voltage, high level (set threshold, low level)	27	Open	22	26	25	21	2.4	4.5	V	9
J2	Q5	Output voltage, high level (set threshold, low level)	34		29	33	32	28				
J3	$\overline{Q}$1	Output voltage, high level (reset threshold, low level)	5		3	6	7	4				
J4	$\overline{Q}$2		12		9	13	14	11				
J5	$\overline{Q}$3		18		16	19	20	17				
J6	$\overline{Q}$4		24		22	25	26	23				
J7	$\overline{Q}$5		31		29	32	33	30				
J8	$\overline{Q}$6		38		36	39	40	37				
J9	$\overline{Q}$4	Output voltage, high level (set threshold, high level)	27	24	Open	25	26	Open				
J10	$\overline{Q}$5	Output voltage, high level (reset threshold, high level)	34	21	Open	32	33	Open				
J11	Q1	Output voltage, high level (reset threshold, high level)	5	Open	3	7	6	4				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE VII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
J12	Q2	Output voltage, high level (reset threshold, high level)	12	Open	9	14	13	11	2.4	4.5	V	9
J13	Q3	↓	18	Open	16	20	19	17	↓	↓	↓	↓
J14	Q4	↓	24	27	Open	26	25	Open	↓	↓	↓	↓
J15	Q5	↓	31	34	Open	33	32	Open	↓	↓	↓	↓
J16	Q6	↓	38	Open	36	40	39	37	↓	↓	↓	↓
J17	$\overline{Q}$4	Output voltage, low level (set threshold, low level)	27	↓	22	25	26	21	0.00	0.40	↓	↓
J18	$\overline{Q}$5	Output voltage, low level (set threshold, low level)	34	↓	29	32	33	28	↓	↓	↓	↓
J19	Q1	Output voltage, low level (reset threshold, low level)	5	↓	3	7	6	4	↓	↓	↓	↓
J20	Q2	↓	12	↓	9	14	13	11	↓	↓	↓	↓
J21	Q3	↓	18	↓	16	20	19	17	↓	↓	↓	↓
J22	Q4	↓	24	↓	22	26	25	23	↓	↓	↓	↓
J23	Q5	↓	31	↓	29	33	32	30	↓	↓	↓	↓

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE VII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
J24	Q6	Output voltage, low level (reset threshold, low level)	38	Open	36	40	39	37	0.00	0.40	V	9
J25	Q4	Output voltage, low level (set threshold, high level)	27	24	Open	26	25	Open				
J26	Q5	Output voltage, low level (set threshold, high level)	34	31	Open	33	32	Open				
J27	$\overline{Q}1$	Output voltage, low level (reset threshold, high level)	5	Open	3	6	7	4				
J28	$\overline{Q}2$		12	Open	9	13	14	11				
J29	$\overline{Q}3$		18	Open	16	19	20	17				
J30	$\overline{Q}4$		24	27	Open	25	26	Open				
J31	$\overline{Q}5$		31	34	Open	32	33	Open				
J32	$\overline{Q}6$		38	Open	36	39	40	37				
K1	Q1	Output voltage, low level (J threshold, low level)	2	4	3	7	6	Open				
K2	Q2	Output voltage, low level (J threshold, low level)	8	11	9	14	13	Open				

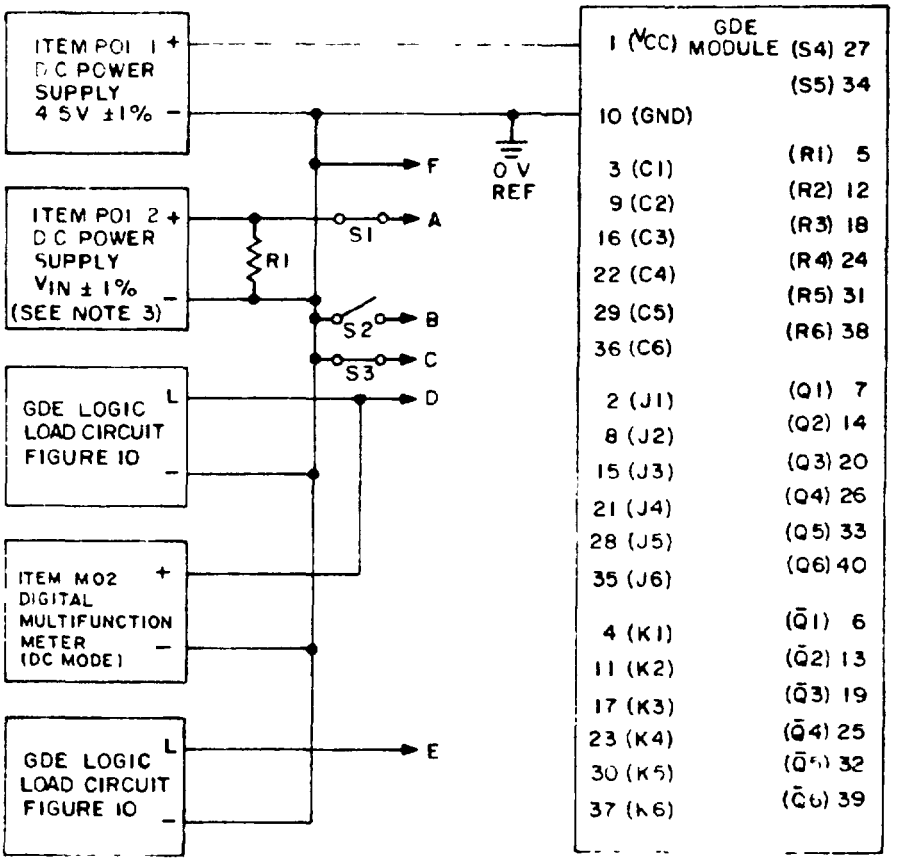
Source: https://assist.dla.mil - Downloaded: 2026-02-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE VII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
K3	Q3	Output voltage, low level (J threshold, low level)	15	17	16	20	19	Open	0 00	0 40	V	9
K4	Q4		21	27	22	26	25					
K5	Q5		28	34	29	33	32					
K6	Q6		35	37	36	40	39					
K7	$\overline{Q}1$	Output voltage, low level (K threshold, low level)	4	5	3	6	7					
K8	$\overline{Q}2$		11	12	9	13	14					
K9	$\overline{Q}3$		17	18	16	19	20					
K10	$\overline{Q}4$		23	24	22	25	26					
K11	$\overline{Q}5$		30	31	29	32	33					
K12	$\overline{Q}6$		37	38	36	39	40					

1/ Test conditions G and H are performed in sequence. Perform test condition G1 without disconnecting the power or point C, setup for condition H1. Perform test condition H1. Repeat this sequence for test conditions G2 and H2 through G6 and H6 The G test conditions set Q in the high level state which is necessary before the H test conditions can be performed.

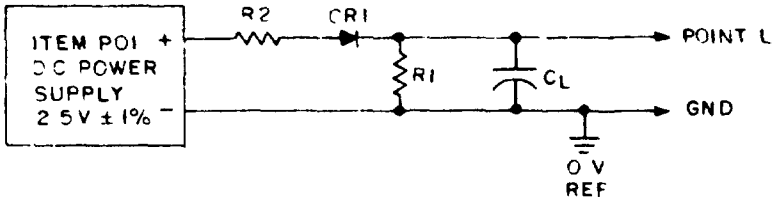
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES
1. R1 = 50Ω ±5%, 1 W
2. S1-S3 = SPST switch
3. Set item P01-2 as follows
   - V_{IN} = 2.0 V ±1% for test conditions G1-G6, H1-H6, J9-J16, and J25-J32
   - V_{IN} = 0.8 ±1% for test conditions J1-J8 and J17-J24
   - V_{IN} = 1.1 V ±1% for test conditions K1-K12
4. Switching sequence for test conditions G, H, J, K4, K5, and K7-K12
   - a. After making all circuit connections, close switch S2 for a minimum of 100 ns, then open it
   - b. Open switch S3 for a minimum of 100 ns, then close and make measurement.
5. Switching sequence for test conditions K1-K3 and K6
   - a. After making all circuit connections, open switch S1 and close switch S2
   - b. Open switch S3 for a minimum of 100 ns, then close it.
   - c. Close switch S1 and open switch S2
   - d. Open switch S3 for a minimum of 100 ns, then close and make measurement

* FIGURE 9. Output voltage (high-level and low-level) test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N4150 or equivalent
2. C_L = 130 ±5 pF (including probe and parasitic capacitance)
3. R1 = 1.91 kΩ ±1%, 1/8 W
4. R2 = Select value for current flow of 22 mA ±1% from point L to ground
5. Capacitor C_L is not required for dc electrical conditions G, H, J and K
6. During output switching, item P01 shall have a maximum overshoot of 500 mV peak

FIGURE 10. GDE logic load circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE VIII. Power supply current (Q low-level) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
L	Power supply current (Q low level)	20	93	mA	11

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:21Z
Check the source to verify that this is the current version before use.

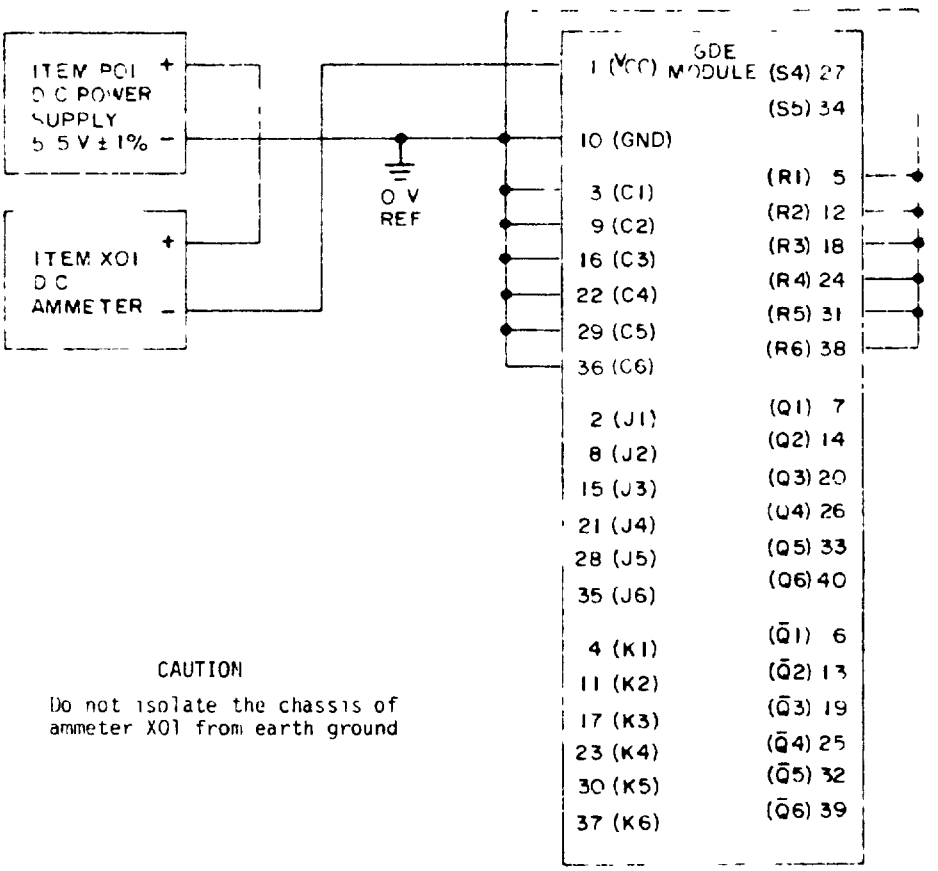


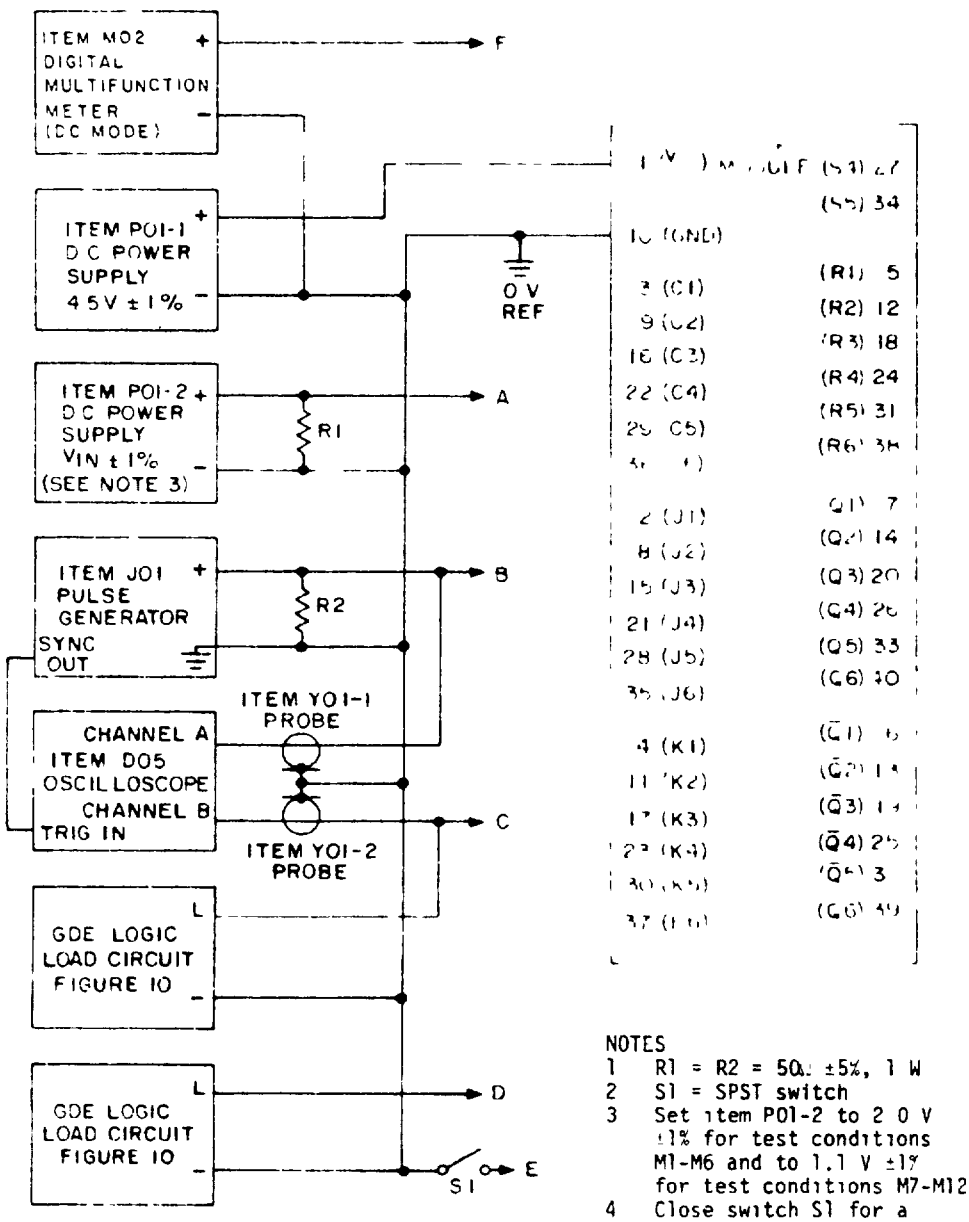
FIGURE 11 Power supply current (Q low-level) test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE IX JK operation (J and K threshold high-level and low-level, maximum clock frequency and minimum pulse width) test

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
1	JK operation (J and K threshold, high level)	2,4	3	7	6	Open	Open	76	84	ns	12,13
2		5,11	9	14	13						
3		15,17	16	20	19						
4		21,23	22	26	25						
M5		28,30	29	33	32						
M6		35,37	36	40	39						
47	JK operation (J and K threshold, low level)	2,4	3	Open	6	5	7	0.3	0.4	V	
M8		6,11	9		13	12	14				
M9		15,17	16		19	18	20				
M10		21,23	22		25	24	26				
M11		28,30	29		32	31	33				
M12		35,37	36		39	38	40				

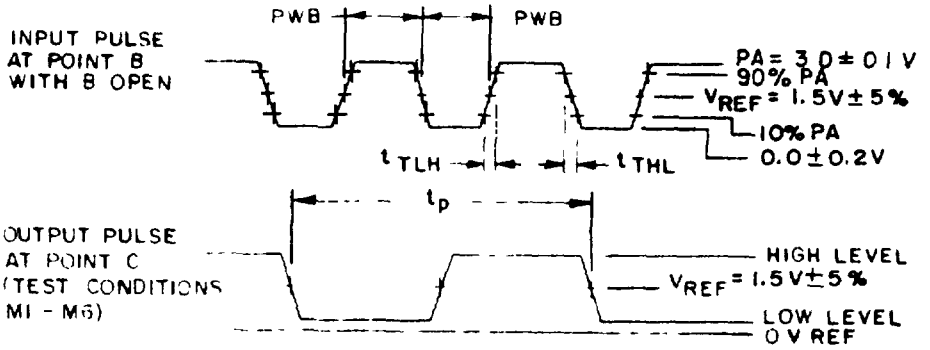
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES
1 R1 = R2 = 50Ω ±5%, 1 W
2 S1 = SPST switch
3 Set item P01-2 to 2 0 V ±1% for test conditions M1-M6 and to 1.1 V ±1% for test conditions M7-M12
4 Close switch S1 for a minimum of 100 ns, then make measurement

FIGURE 12 J K operation (J and K threshold high-level and low-level, maximum clock frequency, minimum pulse width) test circuit for table IX

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:21Z
Check the source to verify that this is the current version before use.



NOTE Item J01 setup
PWB = 20 +2 ns
$t_{TLH} = t_{THL} = 6 \pm 1$ ns
t_p = As specified in table IX

* FIGURE 13 The operation (J and K threshold high-level and low-level, maximum clock frequency, minimum pulse width) test circuit waveforms for table IX

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE X. Transition time (t_{TLH} and t_{THL}) (minimum input setup and hold time) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Q1 Transition time (minimum input setup and hold time) t_{TLH}	2,3,4	7	6	3	20	ns	14,15
N2	Q2	8,9,11	14	13				
N3	Q3	15,16,17	20	19				
N4	Q4	21,22,23	26	25				
N5	Q5	28,29,30	33	32				
N6	Q6	35,36,37	40	39				
N7	$\overline{Q1}$	2,3,4	6	7				
N8	$\overline{Q2}$	8,9,11	13	14				
N9	$\overline{Q3}$	15,16,17	19	20				
N10	$\overline{Q4}$	21,22,23	25	26				
N11	$\overline{Q5}$	28,29,30	32	33				
N12	$\overline{Q6}$	35,36,37	39	40				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE X. Transition time (t_{TLH} and t_{THL}) (minimum input setup and hold time) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Q1 Transition time (minimum input setup and hold time) t_{THL}	2,3,4	7	6	3	20	ns	14,15
P2	Q2	8,9,11	14	13				
P3	Q3	15,16,17	20	19				
P4	Q4	21,22,23	26	25				
P5	Q5	28,29,30	33	32				
P6	Q6	35,36,37	40	39				
P7	$\overline{Q}$1	2,3,4	6	7				
P8	$\overline{Q}$2	8,9,11	13	14				
P9	$\overline{Q}$3	15,16,17	19	20				
P10	$\overline{Q}$4	21,22,23	25	26				
P11	$\overline{Q}$5	28,29,30	32	33				
P12	$\overline{Q}$6	35,36,37	39	40				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) (minimum input setup and hold time) test.

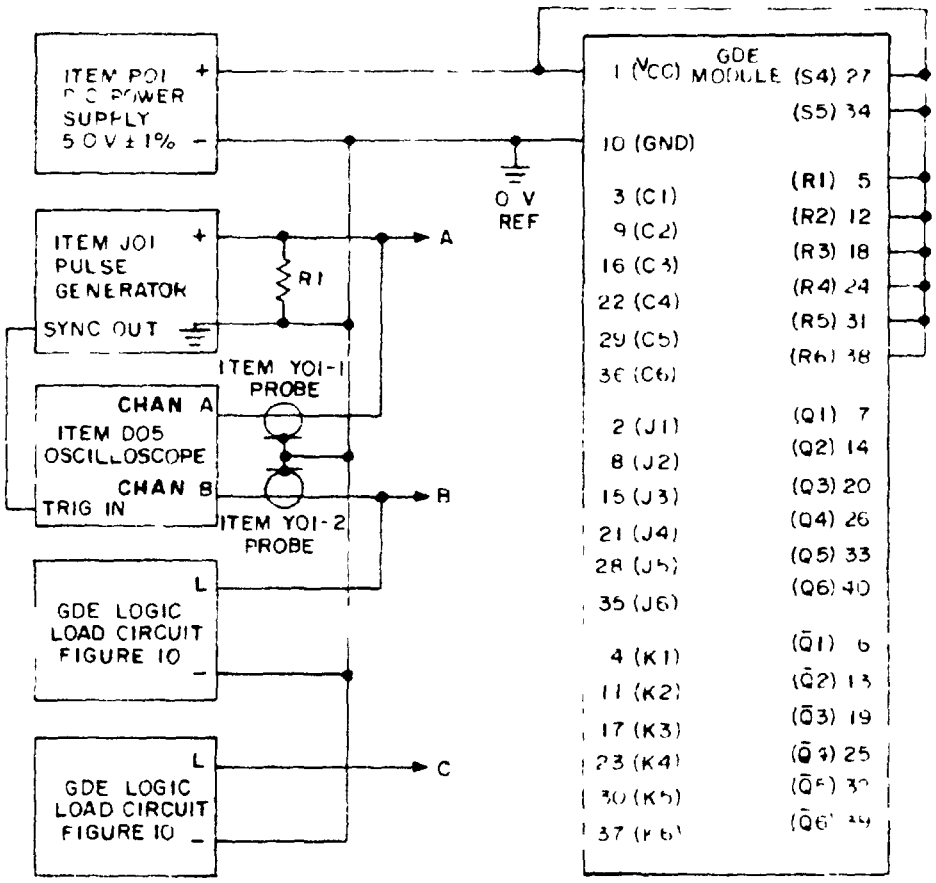
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	Q1 Propagation delay time (minimum input setup and hold time) t_{PLH}	2,3,4	7	6	3	50	ns	14,15
Q2	Q2	8,9,11	14	13				
Q3	Q3	15,16,17	20	19				
Q4	Q4	21,22,23	26	25				
Q5	Q5	28,29,30	33	32				
Q6	Q6	35,36,37	40	39				
Q7	$\bar{Q}1$	2,3,4	6	7				
Q8	$\bar{Q}2$	8,9,11	13	14				
Q9	$\bar{Q}3$	15,16,17	19	20				
Q10	$\bar{Q}4$	21,22,23	25	26				
Q11	$\bar{Q}5$	28,29,30	32	33				
Q12	$\bar{Q}6$	35,36,37	39	40				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE XI Propagation delay time (t_{PLH} and t_{PHL}) (minimum input setup and hold time) test - Continued

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	Q1 Propagation delay time (minimum input setup and hold time) t_{PHL}	2,3,4	7	6	3	30	ns	14,15
R2	Q2	8,9,11	14	13				
R3	Q3	15,16,17	20	19				
R4	Q4	21,22,23	26	25				
R5	Q5	28,29,30	33	32				
R6	Q6	35,36,37	40	39				
R7	$\overline{Q}$1	2,3,4	6	7				
R8	$\overline{Q}$2	8,9,11	13	14				
R9	$\overline{Q}$3	15,16,17	19	20				
R10	$\overline{Q}$4	21,22,23	25	26				
R11	$\overline{Q}$5	28,29,30	32	33				
R12	$\overline{Q}$6	35,36,37	39	40				

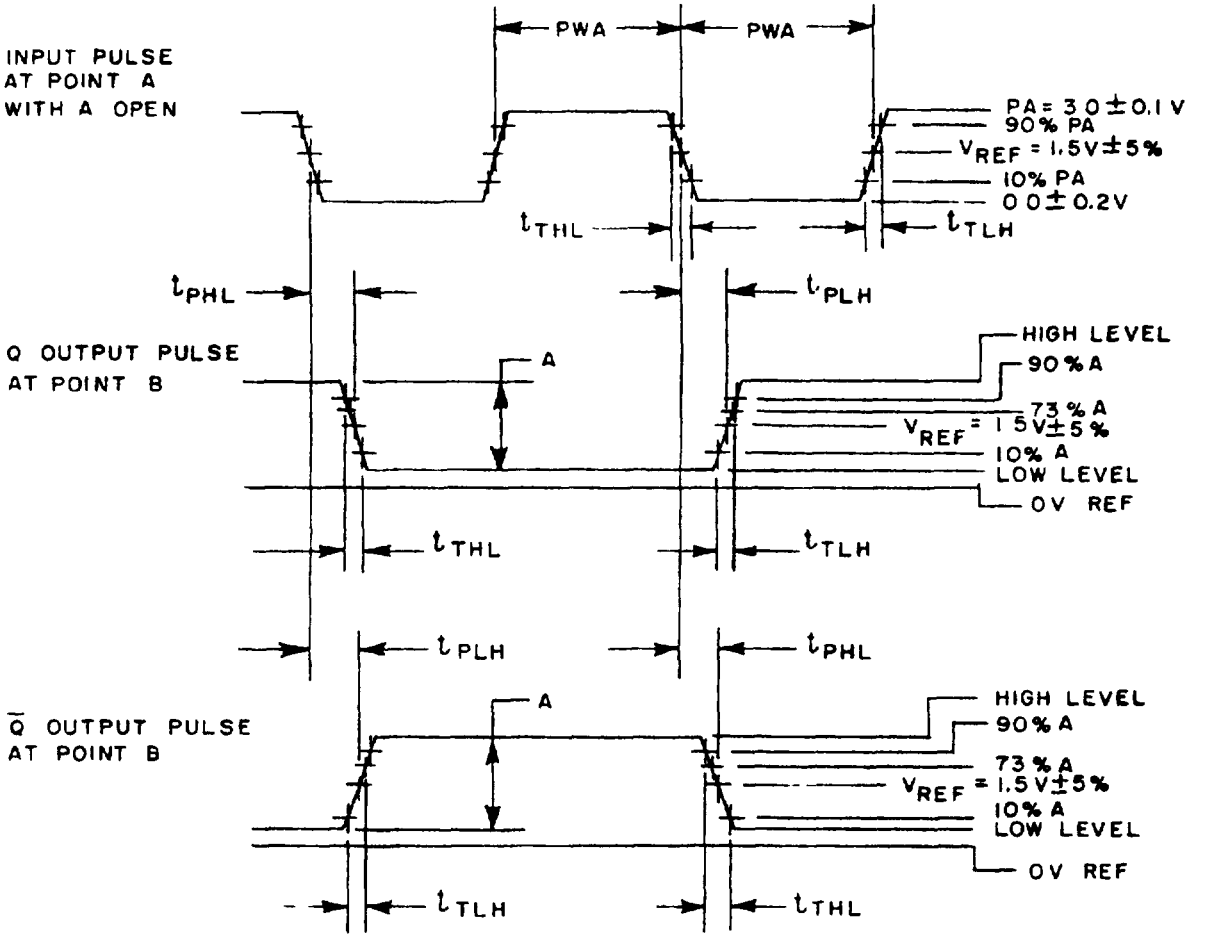
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTE R1 = 50 ±5%, 1 W

FIGURE 14 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) (minimum input set-up and hold time) test circuit for tables X and XI

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters
   PWA = 500 ±50 ns
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns
2. Since the J, K and check inputs are connected together, the requirements of an input setup time of 0 ns and an input hold time of 0 ns is checked
3. Before making the output pulse measurements, the output levels shall be verified to be 0.40 V maximum for low level and 2.4 V minimum for high level

* FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) (minimum input setup and hold time) test circuit waveforms for tables X and XI.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}, set and reset, minimum post time) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	$\overline{Q}4$ Propagation delay time (t_{PHL}), set	22	27	25	26	3	30	ns	16,17
S2	$\overline{Q}5$ Propagation delay time (t_{PHL}), set	29	34	32	33				
T1	Q4 Propagation delay time (t_{PLH}), set	22	27	26	25				
T2	Q5 Propagation delay time (t_{PLH}), set	29	34	33	32				
U1	Q1 Propagation delay time (t_{PHL}), reset	3	5	7	6				
U2	Q2	9	12	14	13				
U3	Q3	16	18	20	19				
U4	Q4	22	24	26	25				
U5	Q5	29	31	33	32				
U6	Q6	36	38	40	39				
V1	$\overline{Q}1$ Propagation delay time (t_{PLH}), reset	3	5	6	7				
V2	$\overline{Q}2$	9	12	13	14				
V3	$\overline{Q}3$	16	18	19	20				
V4	$\overline{Q}4$	22	24	25	26				
V5	$\overline{Q}5$	29	31	32	33				
V6	$\overline{Q}6$	36	38	39	40				

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:21Z
Check the source to verify that this is the current version before use.

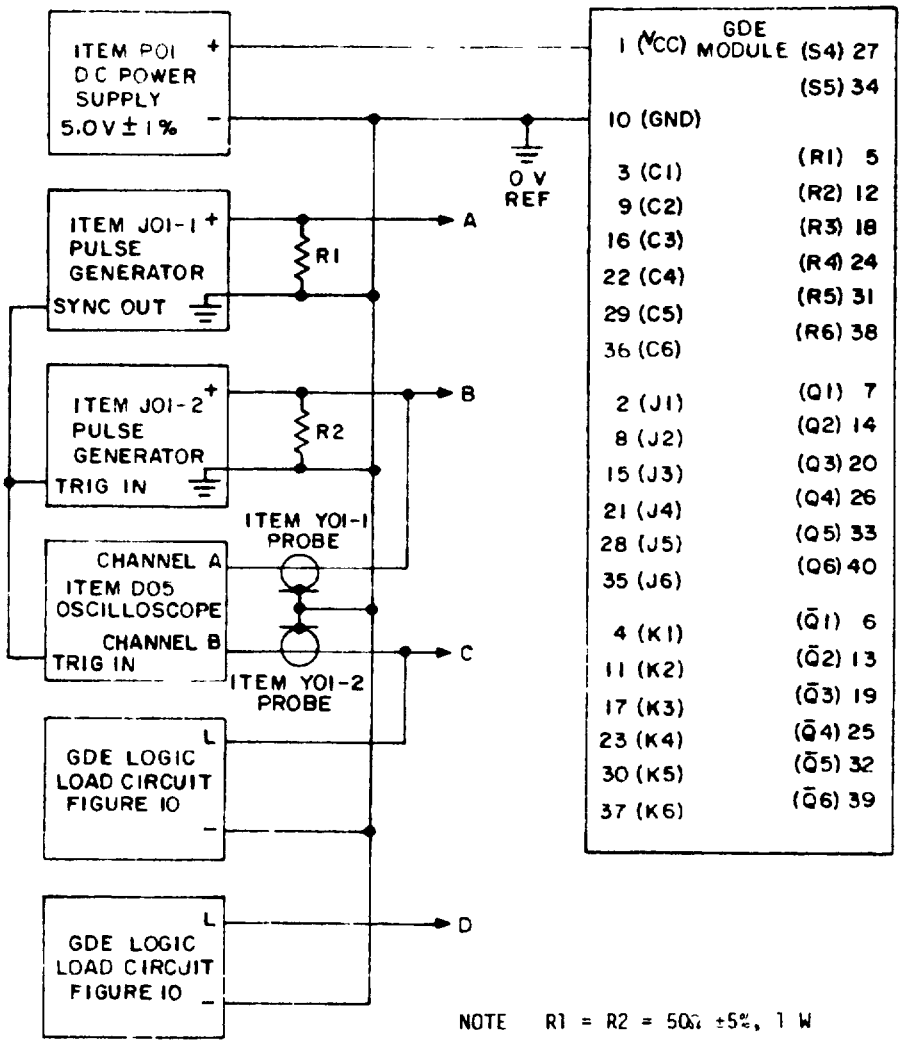
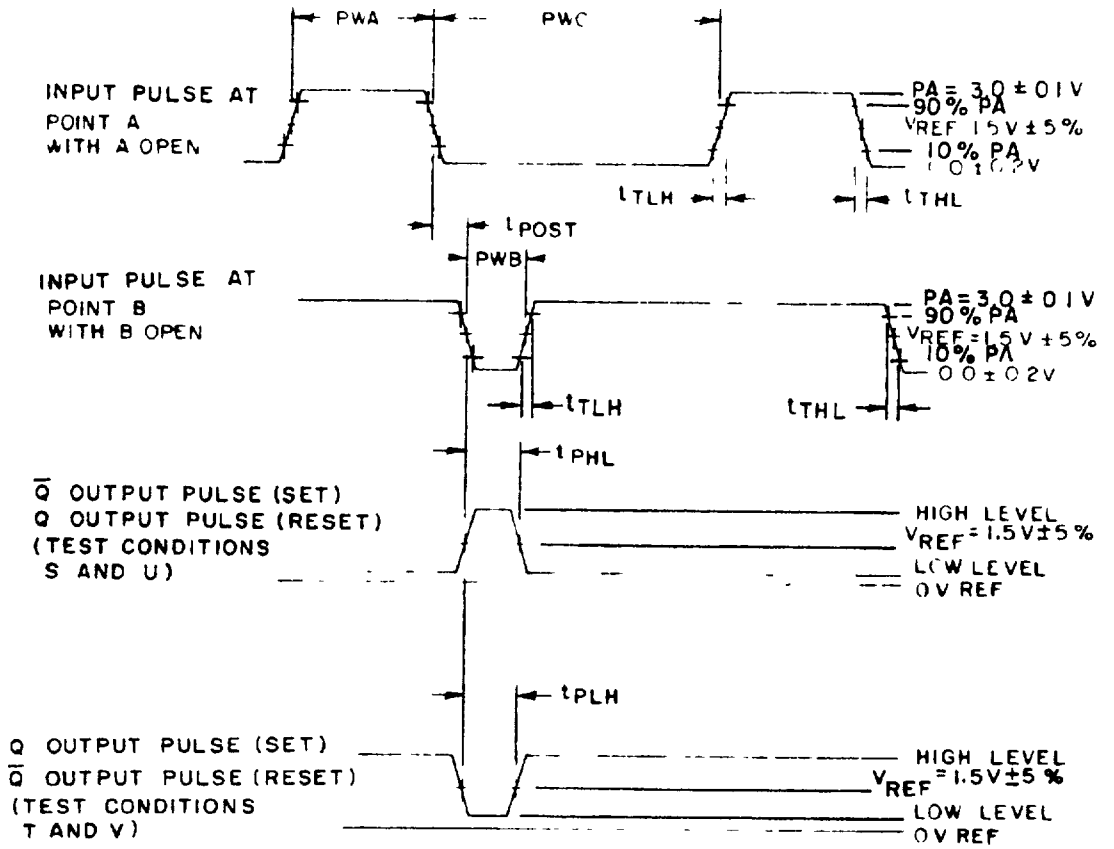


FIGURE 16. Propagation delay time (t_{PLH} and t_{PHL}) (set and reset, minimum post time) test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTE Input pulse parameters
PWA = 100 ±10 ns.
PWB = 20 ±1 ns
PWC = 200 ±20 ns
t_{POST} = 25 ±2 ns
$t_{TLH} = t_{THL}$ = 6 ±1 ns

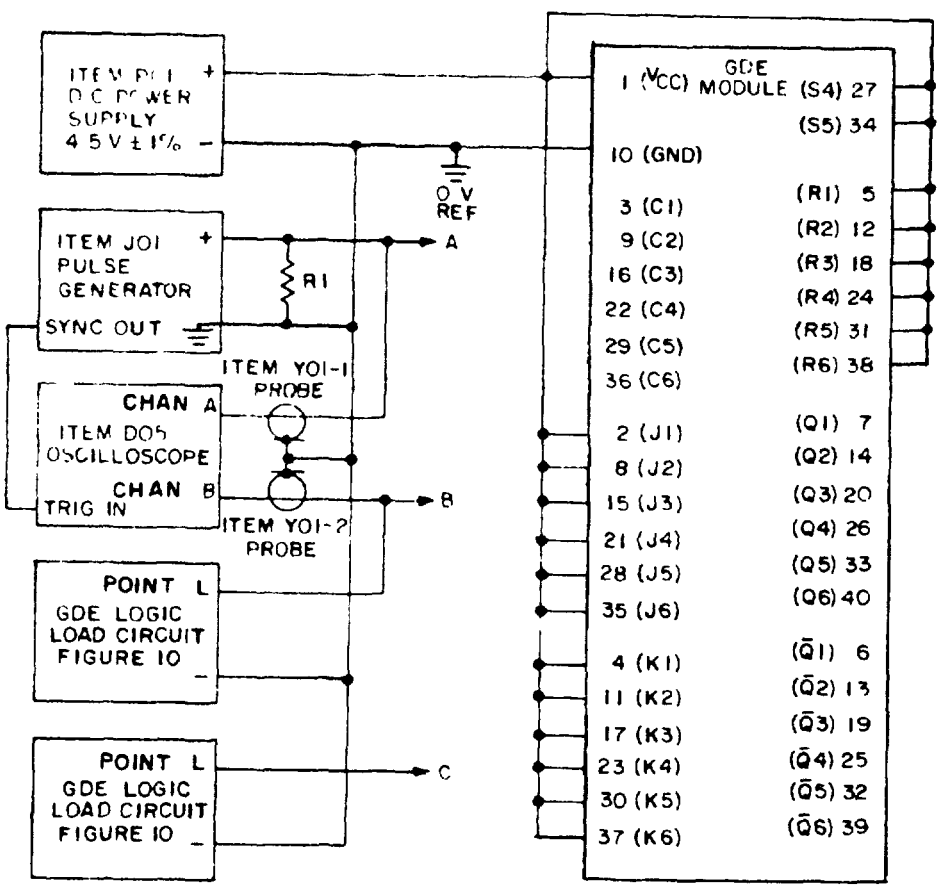
* FIGURE 17. Propagation delay time (t_{PLH} and t_{PHL}) (set and reset, minimum post time) test circuit waveforms for table VII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE XI. Maximum clock fall time and threshold clock input test

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W1	Maximum clock fall time	3	7	6	1.8	2.2	ns	16,19
W2	↓	9	14	13	↓	↓	↓	↓
W3		16	20	19				
W4		22	26	25				
W5		29	33	32				
W6		36	40	39				
X1	Threshold clock input	3	6	7	76	84	ns	18,20
X2	↓	9	13	14	↓	↓	↓	↓
X3		16	19	20				
X4		22	25	26				
X5		29	32	33				
X6		36	39	40				

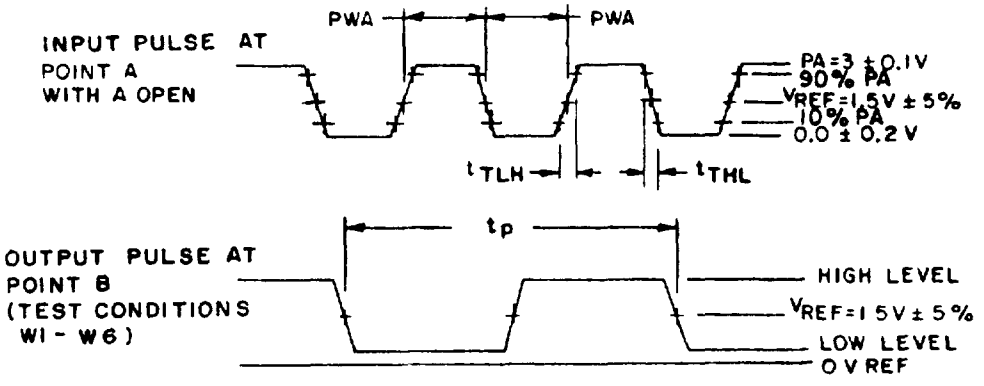
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:21Z
Check the source to verify that this is the current version before use.



NOTE R1 = 50Ω ±5%, 1 W

FIGURE 18. Maximum clock fall time and threshold clock input test circuit for table XIII.

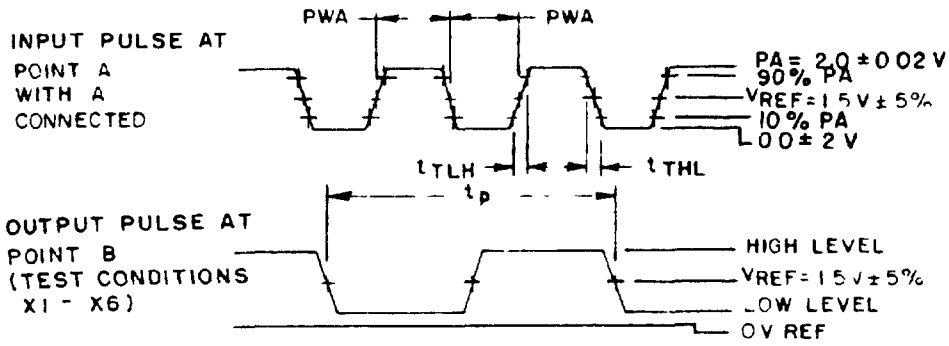
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES
1. Input pulse parameters
   PWA = 500 ±50 ns
   $t_{TLH} = t_{THL} = 100 \pm 5$ ns.
2. t_p = As specified in table XIII.

* FIGURE 19. Maximum clock fall time test circuit waveforms for table XIII.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES
1. Input pulse parameters
   PWA = 20 ±2 ns.
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
2. t_p = As specified in table XIII.
3. All voltage settings are ±50 mV.
4. 0° and 80°C are fin temperatures and 25°C is an ambient temperature.

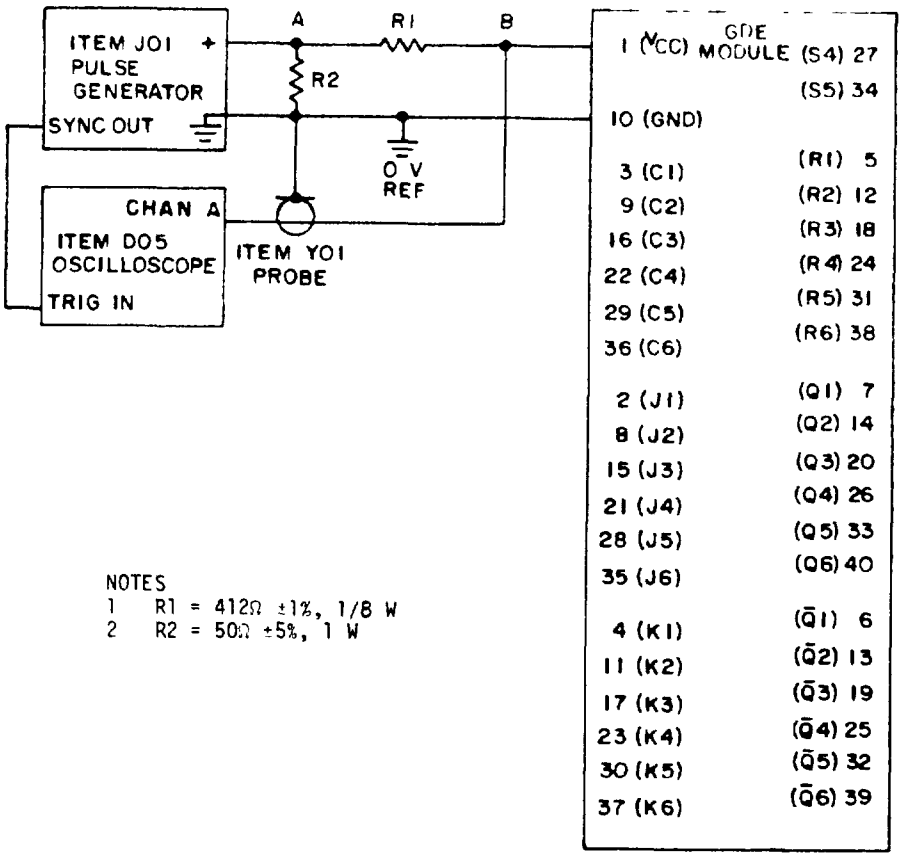
★ FIGURE 20. Threshold clock input test circuit waveforms for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE XIV. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient rise time	2.5	10.0	µs	21,22

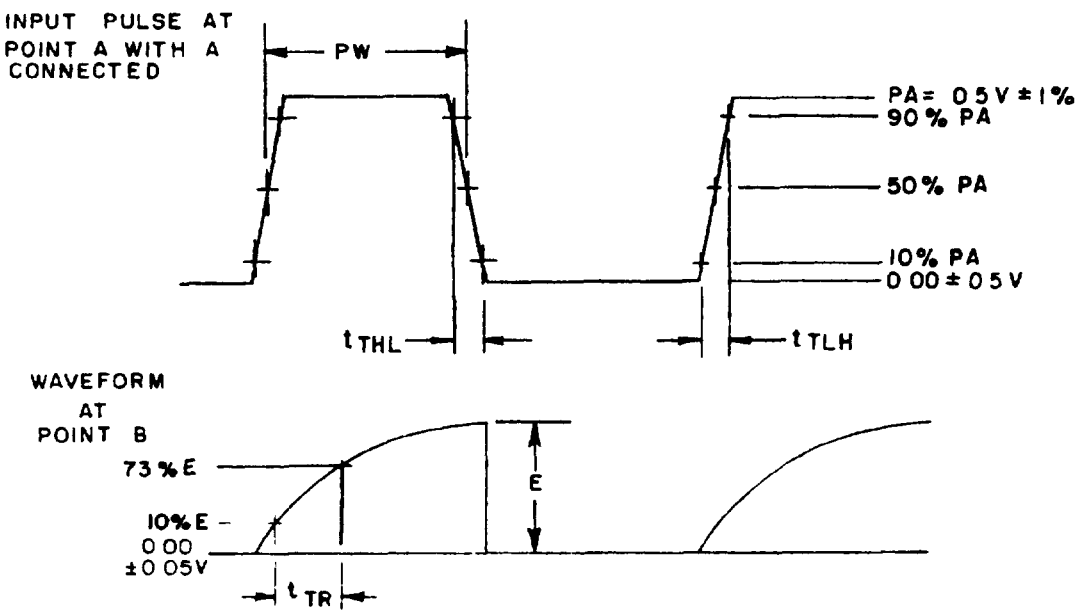
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES
1 R1 = 412Ω ±1%, 1/8 W
2 R2 = 50Ω ±5%, 1 W

FIGURE 21 Power supply transient rise time test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PW = 30 µs ±10%.
   $t_{TLH} = t_{THL}$ = 50 ns ±10%.
   PRP = 60 µs ±10%.
2. Measure transient rise time (t_{TR}) at point B.

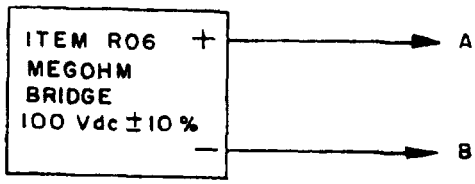
* FIGURE 22. Power supply transient rise time test circuit waveforms for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

* TABLE XV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z	Isolation	All pins	Frame	10	∞	MΩ	23

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:21Z
Check the source to verify that this is the current version before use.



WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

GDE MODULE	
1 (V_{CC})	(S4) 27
	(S5) 34
10 (GND)	
3 (C1)	(R1) 5
9 (C2)	(R2) 12
16 (C3)	(R3) 18
22 (C4)	(R4) 24
29 (C5)	(R5) 31
36 (C6)	(R6) 38
2 (J1)	(Q1) 7
8 (J2)	(Q2) 14
15 (J3)	(Q3) 20
21 (J4)	(Q4) 26
28 (J5)	(Q5) 33
35 (J6)	(Q6) 40
4 (K1)	($\bar{Q}$1) 6
11 (K2)	($\bar{Q}$2) 13
17 (K3)	($\bar{Q}$3) 19
23 (K4)	($\bar{Q}$4) 25
30 (K5)	($\bar{Q}$5) 32
37 (K6)	($\bar{Q}$6) 39

* FIGURE 23. Isolation test circuit for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T15:21Z
Check the source to verify that this is the currentversion before use.

TABLE XVI Output crosstalk (low-level and high-level) test.

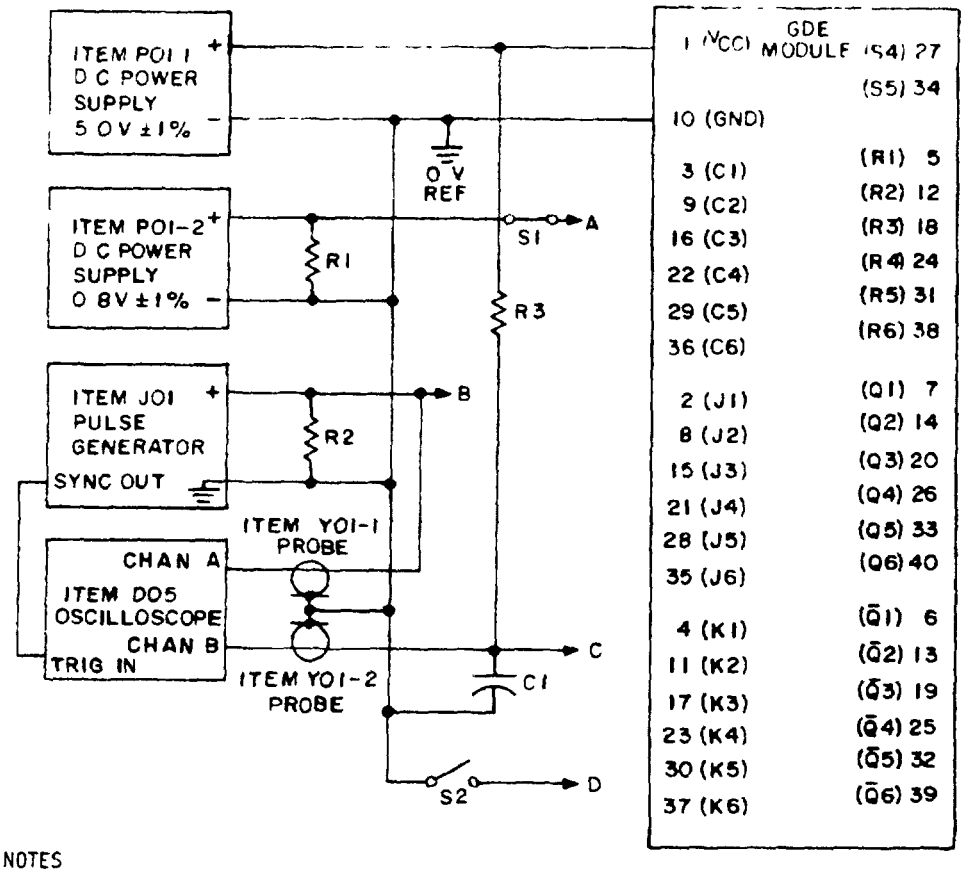
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
AA1	Q1 Output crosstalk, low level	3,5	9,16,22,29, 36	7	12,18,24, 31,38	0	400	mV	24, -
AA2	Q2	9,12	3,16,22,29, 36	14	5,18,24,31, 38				
AA3	Q3	16,18	3,9,22,29,36	20	5,12,24,31, 38				
AA4	Q4	22,24	3,9,16,29,36	26	5,12,18, 31,38				
AA5	Q5	29,31	3,9,16,22,36	33	5,12,18, 24,38				
AA6	Q6	36,38	3,9,16,22,29	40	5,12,18, 24,31				
AA7	$\overline{Q}1$	3	9,16,22,29, 36	6	4,11,17,23, 30,37				
AA8	$\overline{Q}2$	9	3,16,22,29, 36	13	4,11,17,23, 30,37				
AA9	$\overline{Q}3$	16	3,9,22,29,36	19	4,11,17,23, 30,37				
AA10	$\overline{Q}4$	22,27	3,9,16,29,36	25	4,11,17,23, 30,37				
AA11	$\overline{Q}5$	29,34	3,9,16,22,36	32	4,11,17,23, 30,37				
AA12	$\overline{Q}6$	36	3,9,16,22,29	39	4,11,17,23, 30,37				

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
BB1	$\overline{Q}1$ Output crosstalk, high level	3,5	9,16,22,29, 36	6	12,18,24, 31,38	0	400	mV	24,25
BB2	$\overline{Q}2$	9,12	3,16,22,29, 36	13	5,18,24,31, 38				
BB3	$\overline{Q}3$	16,18	3,9,22,29,36	19	5,12,24,31, 38				
BB4	$\overline{Q}4$	22,24	3,9,16,29,36	25	5,12,18,31, 38				
BB5	$\overline{Q}5$	29,31	3,9,16,22,36	32	5,12,18,24, 38				
BB6	$\overline{Q}6$	36,38	3,9,16,22,29	39	5,12,18,24, 31				
BB7	Q1	3	9,16,22,29, 36	7	4,11,17,23, 30,37				
BB8	Q2	9	3,16,22,29, 36	14					
BB9	Q3	16	3,9,22,29,36	20					
BB10	Q4	22,27	3,9,16,29,36	26					
BB11	Q5	29,34	3,9,16,22,36	33					
BB12	Q6	36	3,9,16,22,29	40					

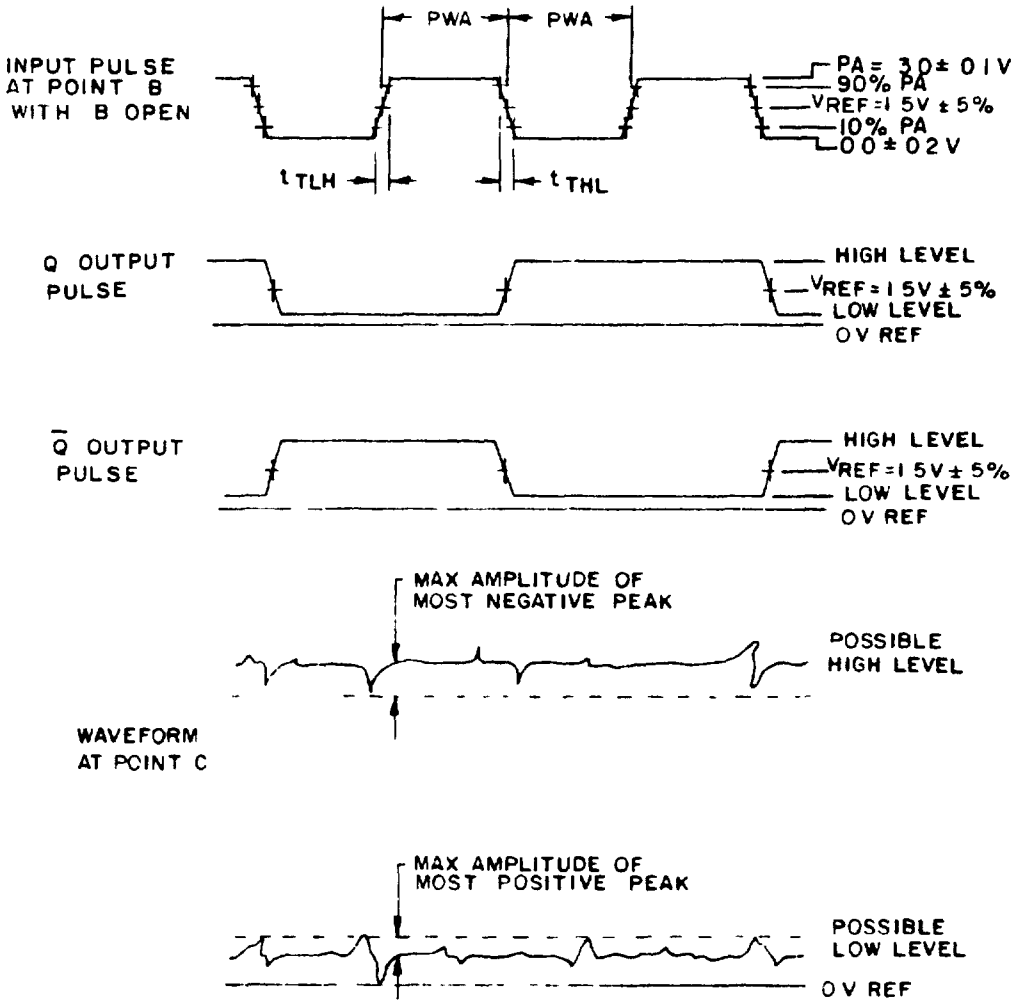
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES
1. R1 = R2 = 50Ω ±5%, 1 W
2. R3 = 243Ω ±1%, 1/4 W
3. C1 = 30 pF ±1% (including probe and parasitic capacitance)
4. S1 = S2 = SPST switch.
5. Only the output under test shall be loaded. The loop length from pin 1 to each load resistance and output, including the length of the resistor body, shall not exceed 4.5 inches (114.3 mm).
6. Switching sequence for test conditions AA1-AA6, AA10, AA11, BB1-BB6, BB10, and BB11.
   a. Switch S1 closed.
   b. Close switch S2 for a minimum of 100 ns, then open and make measurement.
7. Switching sequence for test conditions AA7-AA9, AA12, BB7-BB9, and BB12.
   a. S1 and S2 closed.
   b. Open switch S1 for a minimum of 100 ns, then close it.
   c. Open switch S2 and make measurement.

FIGURE 24. Output crosstalk (low-level and high-level) test circuit for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTE Input pulse parameters
PWA = 500 ±50 ns.
$t_{TLH} = t_{THL} = 25 \pm 5$ ns

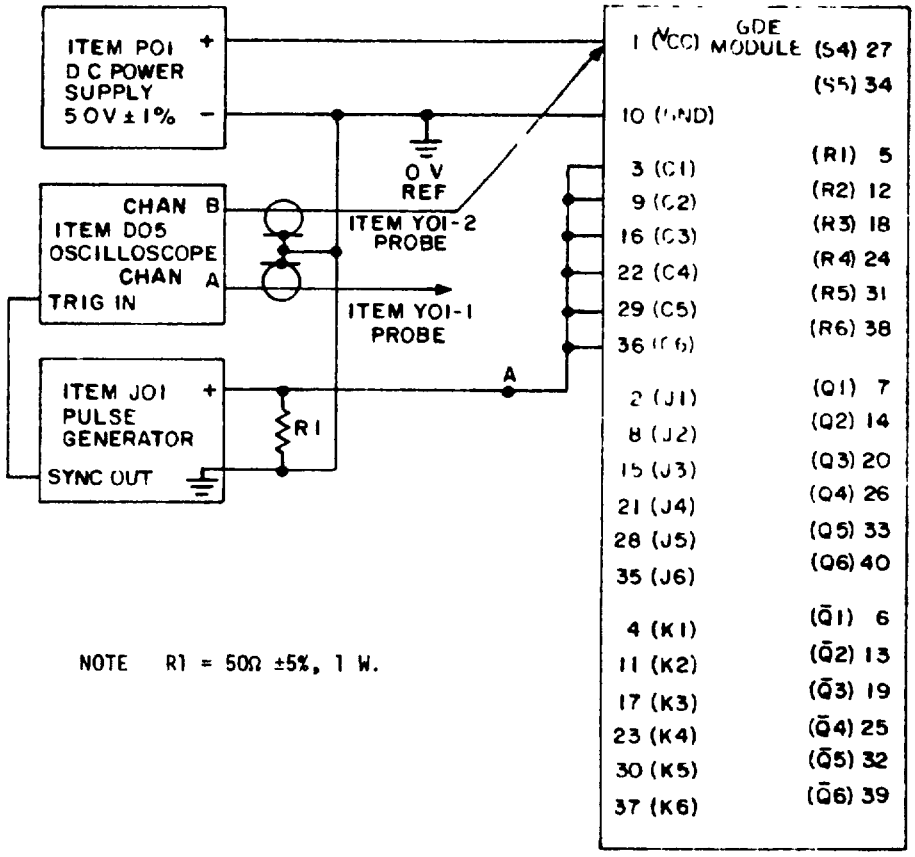
* FIGURE 25. Output crosstalk (low-level and high-level) test circuit waveforms for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE VII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
CC	Power supply transient amplitude	0	400	mV	26,27

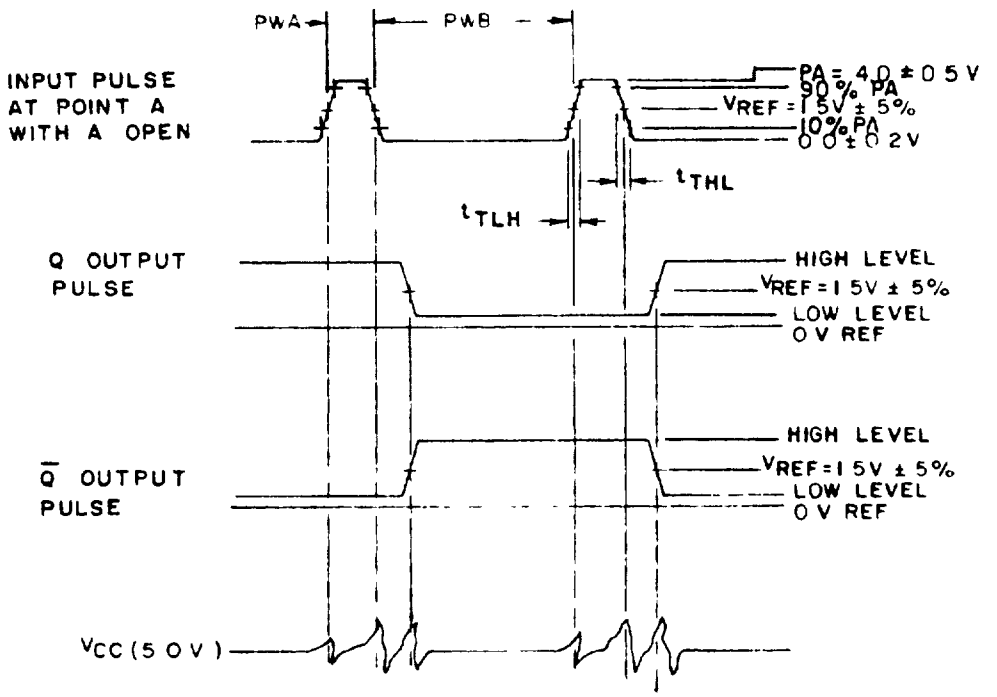
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTE R1 = 50Ω ±5%, 1 W.

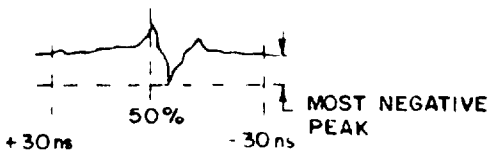
FIGURE 26. Power supply transient amplitude test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES

1 Measurement The most negative peak at the 50% point (+30 ns) must comply with test condition CC



2 Input pulse parameters
PWA = 3 ±0.3 µs
PWB = 7 +0.7 µs
$t_{TLH} = t_{THL} = 10 +2$ ns

* FIGURE 27. Power supply transient amplitude test circuit waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE XVIII Input capacitance tests

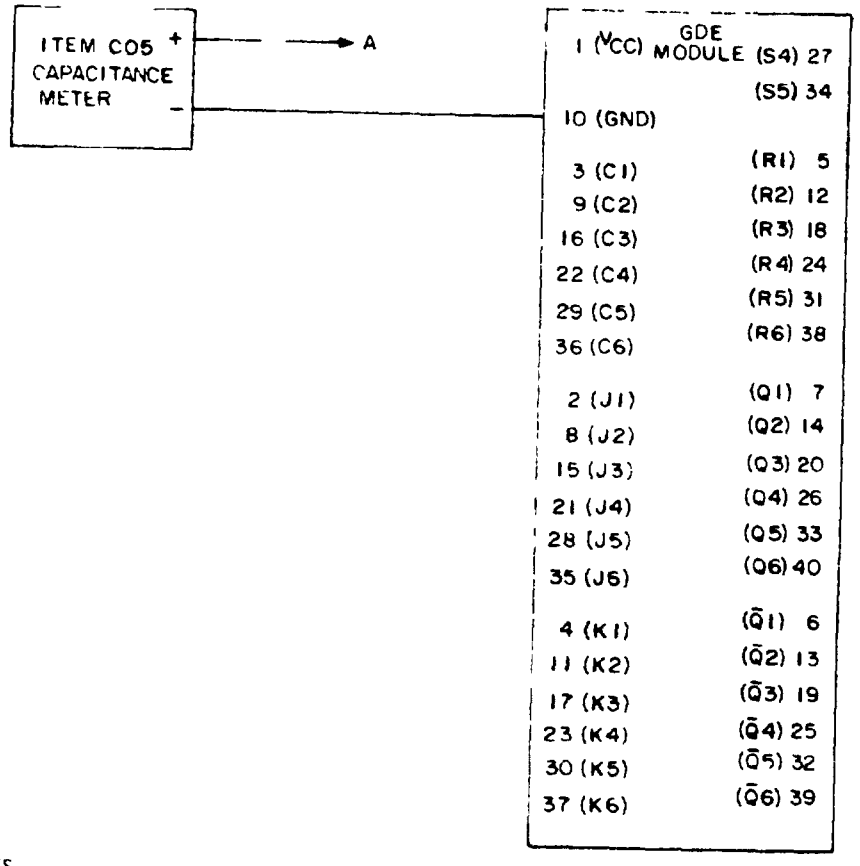
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
DD1	C1	Input capacitance	3	2	15	pF	28
DD2	C2		9				
DD3	C3		16				
DD4	C4		22				
DD5	C5		29				
DD6	C6		36				
DD7	J1		2				
DD8	J2		8				
DD9	J3		15				
DD10	J4		21				
DD11	J5		28				
DD12	J6		35				
DD13	K1		4				
DD14	K2		11				
DD15	K3		17				
DD16	K4		23				
DD17	K5		30				
DD18	K6		37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Input capacitance tests - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
DD19	R1	Input capacitance	5	2	15	pF	28
DD20	R2	↓	12	↓	↓	↓	↓
DD21	R3		18				
DD22	R4		24				
DD23	R5		31				
DD24	R6		38				
DD25	S4		27				
DD26	S5		34				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES
1 Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test Stray capacitance added to other pins may affect the readings
2 Null out the effects of the test leads prior to connecting the module.

FIGURE 28. Input capacitance test circuit for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T15:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-M-28787/2B(EC) SEM, Flip-Flop, JK, Digital, Key Code GDE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.

DLO

PARTMENT OF THE NAVY

OFFICIAL BUSINESS
ENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Electronic Systems Command
Attn: ELEX 8111
Washington, DC 20360

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:21Z
Check the source to verify that this is the current version before use.